Short Project Definition

PitchVision — Broadcast to 2D Mini-Map Projection

Author: Peter Yaacoub
Project Type: Productivity
Course: Computer Vision

---

1 Project Description

1.1 Problem Statement and Motivation

Broadcast cameras provide a great perspective for viewers but fail to show the full tactical layout of a football match. Tactical analysis requires knowing the relative positions of all visible players on a standardized 2D plane (a mini-map). This project will build a pipeline that takes a perspective broadcast view of a football match and projects the visible players onto a simplified 2D pitch map using traditional Computer Vision and Deep Learning techniques.

1.2 Scope and Non-Goals

To ensure the project remains feasible and focused strictly on Computer Vision:

- **I will NOT** attempt to predict future player movements or passes.
- **I will NOT** track players uniquely across camera cuts (tracking will be restricted to continuous camera shots).
- **I will NOT** process 3D ball trajectories (the focus is on 2D floor projections).

1.3 Proposed CV Techniques

Technique	Application / Why I use it
YOLOv8 (Pre-trained)	To detect players, the referee, and the ball rapidly and accurately.
K-Means Color Clustering	Applied to player bounding boxes to separate Team A, Team B, and the Referee based on dominant jersey colors.
Canny Edge & Hough Transform	To detect the white field lines and extract intersection points.
YOLO Feature Map Extraction + Kalman Filter	To perform lightweight frame-to-frame Multi-Object Tracking (MOT). Uses ROI Align on intermediate YOLO layers for appearance matching, a Kalman filter for motion prediction, and the Hungarian algorithm for association.
Homography Matrix (H)	To compute the perspective warp from the camera view to a 2D top-down template using the detected line intersections.
ORB/SIFT + RANSAC	To track background/field features between consecutive frames when the camera pans, allowing smooth updates to the homography matrix.

2 Project Pipeline Design and 50% Milestone

2.1 End-to-End Pipeline

The project is divided into 6 chronological stages:

1. **Player Detection:** Run YOLO to extract bounding boxes for all humans and the ball on the pitch.
2. **Team Classification:** Extract the crop of each bounding box, mask out the green grass background, and use K-Means color clustering to assign labels (Team A, Team B, Ref).
3. **Field Detection:** Apply Canny Edge Detection and Hough Lines to isolate pitch lines and identify key intersections (e.g., penalty box corners, midfield circle).
4. **Consistent Player Tracking:** Extract visual feature embeddings from the detected bounding boxes using YOLO feature maps. Apply a Kalman Filter and the Hungarian Algorithm to assign and maintain unique, consistent tracking IDs across continuous frames.
5. **Homography (Camera Tracking):** Calculate the projection matrix H using the intersections. Use ORB/SIFT + RANSAC to update H continuously during camera panning.
6. **Mini-Map Projection:** Multiply the bottom-center coordinates of each tracked player's bounding box by the Homography matrix to project them onto a 2D radar graphic.

2.2 50% Milestone Definition

(Stages 1, 2, and 3 on static images)
By the mid-point review, the video tracking components will not yet be active. The pipeline must successfully process **single, static frames**.

Deliverables for the 50% checkpoint:

- A working YOLO detection script.
- K-means clustering correctly color-coding players.
- A subset data folder containing 50 labeled frames.
- An intermediate report showing visual qualitative results (bounding boxes + team color labels) on the 50-image subset.

3 Test Design and Image Database

3.1 Image Database & Data Source

The primary data will be sourced from open-source football datasets containing broadcast videos and camera calibration/ground-truth tracking data:

- **SoccerNet-Tracking:** Provides raw broadcast video clips and ground-truth bounding boxes.
- **DFL Bundesliga Data Shootout (Kaggle):** Excellent source of varied broadcast angles and lighting conditions.

3.2 Planned Subset Manifest

Instead of processing full 90-minute videos, I will extract a curated dataset of frames to ensure fast iterations and clean evaluation:

- **50% Milestone Subset:** $\approx$ 50 static, distinct frames (extracted every 10 seconds from 2 different broadcast clips). Used to test YOLO and K-Means.
- **Final Target Dataset:** 5 short video clips (10–15 seconds each, $\approx$ 300 frames per clip) from different stadiums and lighting conditions to test the tracking consistency, RANSAC camera panning, and homography stability.

3.3 Test Design

I will evaluate the system using both quantitative metrics and qualitative visualizations:

Quantitative Metrics:

- **K-Means Classification Accuracy:** F1-score of player team assignments (compared against manual labeling of the static subset).

- **Tracking ID Consistency:** Frequency of identity switches (ID switches) or track fragmentations during continuous shots to evaluate the robustness of the feature-matching and Kalman-filtering framework.
- **Projection Error (Final Phase):** Mean Absolute Error (MAE) measured in meters/pixels between my projected 2D coordinates and the ground-truth SoccerNet 2D coordinates.

Qualitative Evaluation:

- **Mini-Map Video Generation:** A side-by-side video showing the original broadcast on the left (with bounding boxes and unique tracking ID labels) and the live-updating 2D mini-map on the right.